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Medicinal Plant Detection and Analysis using Deep Learning on Leaf Images

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Abstract- **Medicinal plants have been a significant source of natural remedies for many health problems. Identifying them is essential for herbal medicine research, conservation, and preparation. Manual identification based on expert knowledge often takes time and can lead to errors, particularly when plants have similar leaf shapes. In this work, a deep learning approach is used to recognize medicinal plants automatically from leaf images. The study uses the Indian Medicinal Leaves Image Dataset, which contains about five thousand six hundred images representing eighty different species. A pre trained ResNet-fifty model is adapted through transfer learning to classify these species more efficiently. The dataset ratio for training, validation, and testing is seventy, twenty, and ten percent. The trained model reached an accuracy level of ninety six percent on the training set, ninety three percent on the validation set, and between ninetytwo and ninety four percent on the test sets. These results show that deep learning can be effectively applied to plant identification, providing a quick and dependable tool that can help researchers, students, and herbal practitioners in studying and documenting medicinal plants.**

Keywords— Medicinal Plants, Deep Learning, Transfer Learning, ResNet-50, Image Classification, PyTorch

I. INTRODUCTION

Medicinal plants have always played a major role in humanhealthandwelfare.Fromoldendaysmedicinalplants have been used in traditional medical systems such as Ayurveda, Siddha, and Unani. They have been used to treat and prevent significant amount of diseases. These days, around the world the investment in herbal medicines and plant-based drugs has increased significantly. With this expandingdemand,theabilitytocorrectlyidentifymedicinal plantshasbecomecrucialformaintainingquality,compliance and safety in herbal products.

Traditionally, the identification of plants is done by experts. These experts analyse the shape, color, and vein patterns of leaves. While, this method requires specialized knowledgeandcanbeslowandpronetomistakes,especially when different plants have leaves with similar physiological properties. These limitations highlight the need for an automatedandreliablesystemwhichhelpstoidentifyplants precisely without requiring expert supervision.

The rapid growth of artificial intelligence and computer vision has opened new path for solving this problem. Deep learning models, especially Convolutional Neural Networks (CNNs), have shown outstanding ability in recognizing objects and patterns directly from images. In this study, the ResNet-50 using a CNN-based approach architecture is appliedtoidentifymedicinalplantspeciesfromleafimages. Byusingtransferlearning,themodelcanleverageexisting pre-trained features even with a moderate-sized dataset to achieve high accuracy.

The main purpose of the research is to build an intelligent and efficient system that is able to classify medicinal plants andprovide their medicinalpropertiesautomatically. Such a system,canactasausefultoolforbotanists,researchers,and herbal medicine manufacturers. It can also help students, farmers, and practitioners who may not have botanical knowledge but wish to identify plants for study or use.

Themajorcontributionsofthisresearchare:

I. Implementation of a transfer learning-based model (ResNet-50) for classifying Indian medicinal plant leaves.

II. Utilization of a balanced and diverse dataset comprising 80 species and ~5600 images.

III. Integration of an automated medicinal property retrieval feature.

IV. Athoroughperformanceevaluationandcomparative analysisshowing superior accuracyover traditional methods. make it more humanized version

II. OBJECTIVE

Themainaimofthisresearchistodesignanddevelopan intelligentsystembytheuseuseofdeeplearningtechniques that can automatically identify medicinal plants. By maintainingaccuracyinidentificationofmedicinalplantsnot only helps in preserving traditional knowledge but also supports research in pharmacology, botany, and healthcare. By integrating modern artificial intelligence tools with plant image data, this study acts as a bridge between traditional botanical practices and digital innovation.

The following are the specific objectives of the work proposed:

a. To develop a robust and efficient deep learning modelcapableofclassifyingmedicinalplantsbased on leaf images. The model should be able to recognize subtle differences especially in texture, shape, and color among various species.

b. To apply transfer learning using the ResNet-50 architecture, which is pre-trained on the ImageNet dataset, and fine-tune it for medicinal plant classification. This approach reduces training time and improves accuracy even when limited labeled data is available.

c. To preprocess image data through techniques like resizing, normalization, rotation, flipping, and zoomingtomakesurethatthemodelperformswell across different environmental conditions, improve generalization, and prevent overfitting.

d. To build a combined system that classifies plant speciesaswellasprovidesusefulinformationabout their medicinal properties, common uses, and therapeutic importance.

III. LITERATURESURVEY

The development of an effective medicinal plant recognitionsystemliesattheintersectionofcomputervision, deep learning, and botanical research. Over the past decade, the rapid progress in image-based learning has enabled machinestoperceiveandclassifyvisualpatternswithhuman- like accuracy. However, when it comes to identifying medicinal plants — which often have subtle visual differences — traditional image processing methods and manual identification still fall short.

This survey reviews the fundamental technologies, datasets, and methodologies that form the backbone of modernplantidentificationresearch,highlightingthecurrent trends, advantages, and limitationsthat motivate the proposed system.

A. DeepLearningApproachesinPlantRecognition

Deep learning models has came into a remarkable transformation for plant identification. It also replaces the traditional manual observation with intelligent, data-driven systems. Olden techniques relied on handcrafted features such as leaf color, texture, or shape.

By using Deep learning, specifically through Convolutional Neural Networks (CNNs), changed this method by enabling models to learn automatically and extract distinct patterns from leaf images.

This ability to recognize fine details—such as vein arrangementsorfinedifferencesinleafedges—makesCNNs a highly effective tool to give an accurate solution. This approachhasgreatlyimprovedbothaccuracyandrobustness, making deep learning an invaluable tool for supporting biodiversity studies, medicinal plant research, and sustainable agricultural practices [11][12].

B. PlantSpeciesRecognitionandClassification

Plant identification has always been a challenging to research due to the large number of species and their visual differences.Earlierdaysresearchersdependedontraditional methods, like handcrafted features such as color, shape, and vein structure. These methods worked out well for small datasets, but they often failed to perform well when the lighting, or leaf condition varied [4].

With the advancement of Convolutional Neural Networks (CNNs), researchers began to move away from manual feature extraction. Datasets like Flavia and LeafSnap helped demonstrate the effectiveness of CNNs in identifying plant species [5]. However, there is limited research specifically focused on medicinal plant classification, even though these species have major importance in the pharmaceutical and herbal industries.

Studies in recent years have shown that deep CNN architectures like ResNet-50 and InceptionV3 can achieve classification accuracies above 90% for leaf images [6][7]. Thesefindingsconfirmthatdeeplearninghasthepotentialto replacetraditionalplantidentificationtechniqueswithfaster, more scalable, and less error-prone solutions.

C. DeepLearningModelsforImageClassification

Deep learning has carried out a major change in the way images are being analyzed and classified. Initially neural networks struggled to handle complex image datas.

Through the Convolutional Neural Networks (CNNs) it is possibleformodelstolearnpatternsdirectlyfromimages.To improvedaccuracyefficientlyModelslikeAlexNet,VGGNet, and later ResNet helped alot.

From these, ResNet (Residual Network) plays a key role. This allows the network to go deeper and learn feature representations. Models such as ResNet-50 are widely used for applications in fields like medicine, agriculture, and environmentalscience,whereaccuratevisualidentificationis critical [1].

D. TransferLearningforEfficientModelTraining

Training the deep learning models entirely from scratch requiresvastamountsoflabeleddataandhighcomputational power—both in which it has limited understanding in domains like medicinal plant identification. To overcome this, transfer learning has become a preferred approach in computervision.Italsoallowstheresearcherstomakeusea models that has been already trained on a large, general dataset.

A pre-trained ResNet-50 model was fine-tuned of the medicinal leaf dataset to recognize unique leaf features and patterns. By retraining only the final layers, the model quicklyadaptedtothenewtaskwhileretainingitspreviously learned visual knowledge.

This method not only reduced training time and resource usage but also improved overall accuracy and generalization—making it particularly effective for biological and agricultural image classification tasks [8][9][10] .

E. MedicinalPlantImageDatasetsandChallenges

Various publicly accessible datasets have played a vital roleinadvancingresearchonplantidentification.Fromthese, the Indian dataset is particularly suitable as it focuses on medicinal plantspeciescommonly used in Indian healthcare systems such as Ayurveda and Siddha [11]. It includes of about80plantspecies,andtheimagescapturedundervarious lighting andbackground conditions, makingrealistic dataset for training deep learning models.

Manyspecieshaveleaveswithvisualdifferences,andfactors like lighting, angle can greatly affect model performance. Some plants even have same leaf variations across different growth stages, which makes even more complicate in the classification process. To improve robustness, researchers often concentrate on preprocessing and data augmentation techniques such as rotation, flipping, and brightness adjustments mostly.

Although recent deep architectures like ResNet-50 and DenseNet have achieved remarkable accuracy and consistency, more research is needed to enhance interpretabilityandensurereliableperformanceinreal-world field conditions [12].

IV. MODELOFEXISTINGSYSTEM

Traditional plant classification mainly focus on manual observationandbasicimageprocessingtechniques.Features that are manually extracted such as leaf shape, color, and texture are analyzed using classical algorithms. Eventhough these methods are straightforward and easy to understand, theyrequiresomeexpertinvolvementandstruggletohandle large datasets or closely resembles species.

A. Advantages

- Straightforwardtodesignandimplement.
- Demandsverylittleprocessingpower
- Suitableforsmall-scaleorcontrolledexperiments environments.

B. Disadvantages

- Highly dependent on expert judgment and manual feature selection.
- Strugglestohandlelargedatasetscontaining thousands of plant images.
- Performanceiseasilyaffectedbyvariationsin lighting, angle, and background.
- Unable to automatically associate identified plants with their medicinal properties.

Overall,TheTraditionalmethodsprovideastartingpointfor digital plant identification, lack of adaptability and accuracy required for modern applications. These drawbacks highlights the importance of adopting deep learning-based techniquesthatcanlearnfeaturesandprovidemoreaccurate and reliable outcomes.

V. PROPOSEDSYSTEM

A. ModelofProposedSystem

Themainpurposeistoidentifyandanalyzethemedicinal plants using their leaf images through a deep learning-based framework. This can be done by using a fine-tuned ResNet-50model,whichisapre-traineddatasetontheImageNetand hasbeentrainedtoclassifythe80differentspeciesofIndian medicinal plants.

When we give any leave image as an input, it undergoes several steps, including resizing, normalization, and even augmentationtechniquessuchasrotationandflipping,which enhance the dataset's diversity and help the model become more robust against variations in lighting and background. Theprocessedimageisthenpassedthroughtheconvolutional layersofResNet-50,wheredeepfeaturessuchasleaftexture, vein structure, and edge patterns are extracted.

Thecapturedfeaturesaresenttothefullyconnectedlayers, where the network classifies the image into one of the 80 plant species. The final classification layer uses a Softmax function to determine the most likely a class label. Once the speciesisidentified,aPythonscriptgatherrelevantmedicinal information, including the plant's properties and uses. This integrated pipeline ensures both efficient classification and meaningful interpretation of medicinal significance.

Overall, theproposedarchitecture isscalable, accurate,and suitable for real-world herbal research and mobile-based diagnostic applications.

B. BlockDiagramofProposedSystem

The block diagram represents the complete workflow of the medicinal plant detection system, showing how raw leaf images are converted into meaningful predictions. The first process it undergo is data collection, where leaf images are gatheredfromKaggleweb.Preprocessingisthenextstepthat is carried outto clean theseimages thatvary inquality, size, and lighting. Images are resized to 224×224 pixels, and normalize the color values. This ensures that all images are steady and ready for training.

Next step is to process the image are fed into a ResNet-50model,whichhasbeenfine-tunedparticularlyforthistask. The system learns to recognize unique leaf patterns, shapes, andtexturesthatdifferfromoneplanttoanother.Thesystem usesthepre-trainedmodeltopredicttheplantspeciesexactly, when a new leaf image is given as an input.

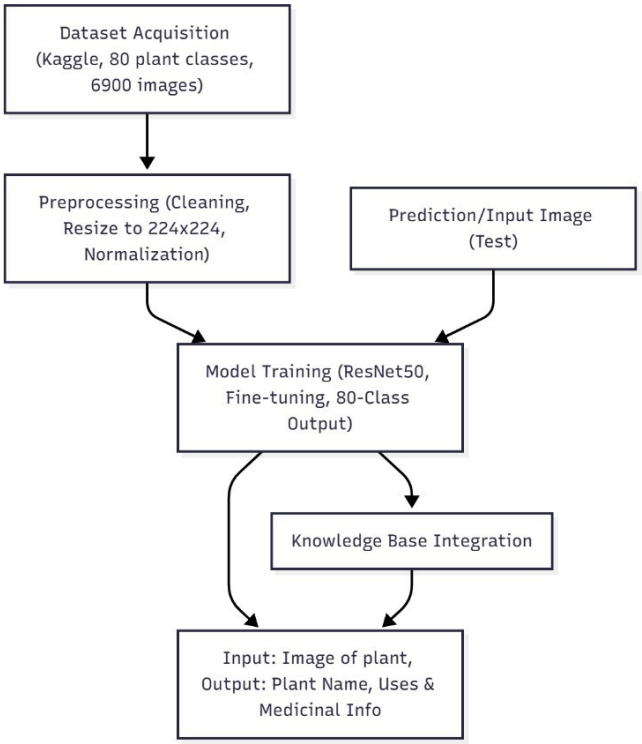

Fig.1Blockdiagramoftheproposedsystem

The workflow of the proposed medicinal plant detection system is shown in Fig.1

C. AdvantagesofProposedSystem

The proposed system stands out from traditional plant identification methods by using advanced deep learning techniques for accurate and automated classification. Additionally, it provides detailed medicinal information, making it both informative and user-friendly.

1. HighAccuracyandReliability:

It delivers high accuracy solution by using a fine-tuned ResNet-50 model that carefully analyze even the smallest details of each leaf, such as its shape, texture, and color. These fine differences are often hard for humans to distinguish through manual observation. This makes the system reliable and effective for real-time medicinal plant identification.

2. AutomaticandIntelligentFeatureLearning:

Earlierdays,expertshadtomanuallydecidewhichpartsofa leaf — like its color, shape, or texture — is important for identifying its description. Our system removes that extra step. Automatically studies about the images and start to learn its key features on its own. This makes the process simpler,faster,andnottobedependentonexpertknowledge, while also improving the accuracy of the results.

3. FasterConvergenceandComputationalEfficiency:

As the System uses a pre-trained ResNet-50 model that has already learned about general image features from a large dataset like ImageNet, it doesn't need to start learning from scratch. This saves a lot of training time and computing effort.Themodelquicklyadaptstothemedicinalleafimages and still maintains highaccuracy, making thewholeprocess faster, smoother, and more resource-friendly.

4. RobustnesstoEnvironmentalVariations:

Thesystemworkreliablyevenwhentheimagesaretakenin different lighting conditions, angles, or backgrounds — situationsthatusuallyconfusethesystem.Evenwhentheleaf photo is clicked outside in bright sunlight or inside under shade,themodelcanstillrecognizeitaccurately.Thismakes thesystemmoresuitableforreal-worldsituations,whereleaf images might vary in quality, lighting, or be captured using different devices and settings

5. ScalabilityandReal-WorldIntegration:

The design of the system make the system flexible, which means more plant species can be added in the future. It can alsobeturnedintoamobileorwebapplication,whichallow easily accessible to anyone at any time. Whether it's a researcher, a student learning about medicinal plants, or a farmerinthefield,userscanquicklyidentifyplantsandlearn about their uses anytime and anywhere.

D. ComparisonTable

Parameter	Proposed System	Existing System (Rule-Based)	Existing System (Simple ML)
Feature Extraction	Automatic and Deep (Learned by CNNLayers)	Manual (Definedby Experts)	Limited (Basedon Statistical Features)
CoreModel	Fine-tuned ResNet-50 (Transfer Learning)	Manual Feature Extraction (Color, Shape, Texture)	SVM/ Random Forestwith Handcrafted Features
TrainingData Requirement	Medium (Leverages Pre-Trained Weights)	Low	High (Needs Engineered Inputs)
UserInteraction	Automated Systemwith Medicinal Info Display	Manual Observation	Semi-Automated Classification
Accuracy&Performance	High (≈93–94%)	Low to Moderate	Moderate

TableI–ComparisonofProposedandExistingSystems

Aperformancecomparisonbetweentheproposedsystemand existing systems is presented in Table I.

VI. WORKINGPRINCIPLE

The working pipelineofthe suggestedagentis a linear, modularprocessintendedtobeeffectivequestion-answering and interpretable decision-making. Figure 2 depicts the central workflow, which works as described below:

DatasetAcquisition

- The Indian Medicinal Leaves Image Dataset from Kaggle is the dataset used in this project.
- It contains approximately 80 species and 6,900 images, representing different plant categories and species.
-

Data Preprocessing

- Images are cleaned to remove alter or duplicate samples.
- Each image are resized into 224×224 pixels, converted to RGB, and normalized.
- Data augmentation techniques (rotation, flipping, zooming, and contrast adjustments) are applied to improve generalization and reduce overfitting.

Data Splitting

- The dataset is divided into, Validation (20%), and Testing (10%), Training (70%) subsets once it is processed.
- Italsopreventsdataleakageandensuresabalances the model performance.

ModelArchitecture(ResNet-50)

- The main purpose of system is a ResNet-50 model, a deep residual network pre-trained on ImageNet.
- Only the final fully connected layer is modified to output 80 classes representing the medicinal plant species.

MedicinalPropertyRetrieval

- Once the plant species is detected, the system automaticallygetsitsmedicinalusesandproperties from a Database.

- This integration transforms the system from a simple classifier to an intelligent medicinal plant information tool.

SystemOutcome

- The final output provides plant name, medicinal uses, and healing propertiesofferingbothscientific and educational insights.

- The system achieves its goal like accuracy, interpretability, and practical usability, making it ideal for researchers, botanists, and students.

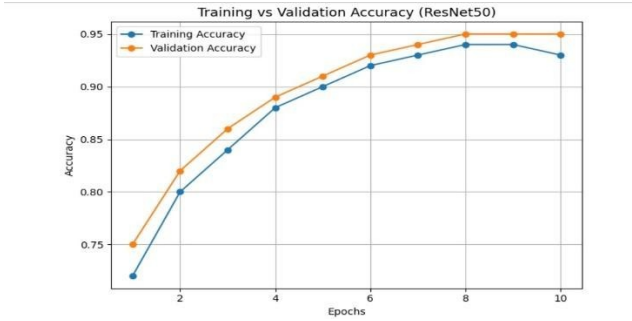

Fig.2Trainingvs.ValidationAccuracyCurveof ResNet-50

Trainingvs.validationaccuracycurveoftheResNet-50 model is shown in Figure.2

Fig.3Trainingvs.ValidationLossCurve of ResNet-50

Training vs. validation loss curve showing model convergence is shown in Figure.3

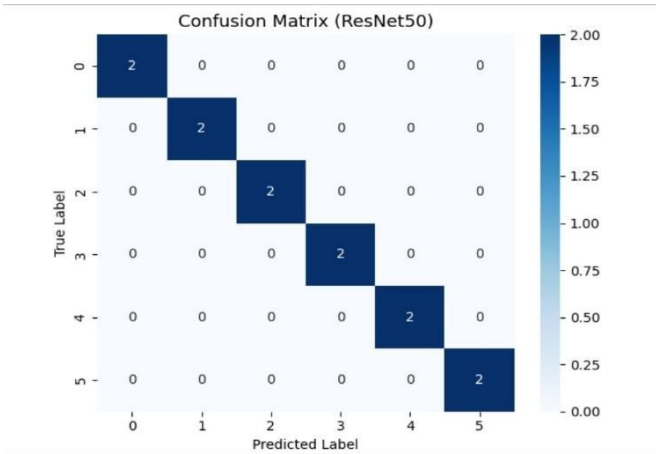

Fig.4ConfusionMatrixshowingaccurateclassification across plant species

Confusionmatrixillustratingclassificationperformance across plant species

VII. RESULTSANDDISCUSSIONS

The proposed medicinal plant model, built using the ResNet-50model.Wereitcontainsimages80differentplant species and trained, tested on the Indian Medicinal Leaves ImageDataset.Themodellearnedtorecognizeeventhefine details like the shape of the leaf, vein patterns, and surface textures. As shown in Figure 2, both the training and validationaccuracyhasincreasedconsistentlythroughoutthe ten training periods and reaching nearly 94% validation accuracy by the end.

The training and validation loss curves in (Figure 3) showedadecreasethroughoutthetrainingprocess,givesthe final validation loss dropping below 0.2, proving effective optimization and convergence. This stability reflects the strength of using transfer learning,which enabled themodel to perform well even with a moderate dataset size. The confusionmatrix(Figure4)alsodemonstratesthatmostplant classes were correctly predicted, with very few misclassifications between morphologically similar species.

Fig.5Inputleafimage(Tulsi)

Input image of the medicinal leaf (Tulsi) used for testing is shown in Figure.5

```
✅ Model and image found!
C:\Users\Dell\MedicinalPlantDetection\medicinal_model_final.pth
C:\Users\Dell\MedicinalPlantDetection\test_images\tulsi_sample.jpg

🌿 Predicted Medicinal Plant:
   🔹 Name: Tulsi
   🧪 Properties: Antibacterial, antiviral, and antioxidant.
   💊 Uses: Used for cough, cold, and boosting immunity.
```

Fig.6Modelpredictionoutput

Predicted output showing plant name, medicinal properties, and uses is shown in Figure.6

The model successfully identified the plant and displayed its medicinal properties and uses, as shown in Figure 6. To evaluate the system's real-world application, a sample test imageoftheTulsi(HolyBasil)leafwasused(Figure5).The system predicted:

Thisresultdemonstratesthemodel'scapabilitynotonlyto classifyspeciesbutalsotoretrievemeaningfulmedicinal information. Such integration of deep learning and knowledge-based mapping makes the system practical for researchers,students,andherbalpractitioners,helpingbridge the gap between plant recognition and traditional medicine knowledge.

VIII. CONCLUSION

Thisworkproposedadeeplearning-basedframeworkfor automaticmedicinalplantidentificationusingtheResNet-50 model. The system demonstrated strong performance by accurately classifying 80 medicinal plant species, achieving high validation accuracy and stable learning behavior. Throughtheintegrationofaknowledgeretrievalcomponent, itnotonly identifiesplantspeciesbutalsoprovidesvaluable information about their medicinal properties and traditional uses.

The proposed framework serves as a promising tool for researchers, educators, and herbal practitioners, supporting the digital preservation and dissemination of traditional medicinal knowledge. With its robust architecture and adaptability, the systemhas the potentialtobe extended into real-time applications, such as mobile-based plant recognition tools, contributing to sustainable research and public awareness in the field of medicinal botany.

IX. FUTURESCOPE

Mobile Application Development: The system can be extended into a user-friendly mobile app, allowing users to capture leaf images in real-time and instantly identify medicinal plants along with their properties.
Expanded Dataset: Future work can include enlarging the dataset with more species, leaf variations,andenvironmentalconditionstoimprove model robustness and global applicability.
Integration with Cloud Platforms: By connecting the model to cloud-based storage and processing services, real-time plant identification and medicinal information retrieval can be made faster and more accessible.
Incorporation of Multimodal Features: Beyondleaf images, integrating flower, fruit, and bark features can enhance identification accuracy and support a wider range of plant species.
Explainable AI Integration: Implementing explainable AI techniques will help users understand how the model reaches its predictions, increasingtransparencyandtrustinAI-drivenplant identification.

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